Probit Regression

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Introduction

Probit regression models a binary outcome through the inverse of the standard Normal cumulative distribution function — the same idea as logistic regression but with a Normal-tailed link instead of a logistic-tailed one. The two methods produce essentially identical predicted probabilities; their coefficients differ by a fixed scaling factor of about 1.6. Probit is the natural choice when a latent Normal-variable interpretation matches the substantive theory; logistic is more common in clinical reporting because odds ratios are easier to communicate.

Prerequisites

A working understanding of logistic regression, the standard Normal CDF, and the latent-variable interpretation of binary regression.

Theory

The probit model is

$$P(Y = 1 \mid X) = \Phi(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p),$$

where Φ is the standard Normal CDF. Coefficients are interpretable on the probit scale: a one-unit increase in X_j corresponds to a β_j-standard-deviation change in the underlying latent Normal propensity. The inverse relationship $\Phi^{-1}(p) = X^\top\beta$ is the probit link.

Probit coefficients are approximately logistic coefficients divided by 1.6 ($\sigma_{\text{logistic}} = \pi/\sqrt{3} \approx 1.81$ vs. $\sigma_{\text{Normal}} = 1$). The link functions agree near $p = 0.5$ but diverge slightly in the tails — logistic has heavier tails — so coefficient comparisons across the two scales require the rescaling.

Assumptions

Binary outcome, independent observations, latent linear relationship on the probit scale, and adequate sample size relative to predictors (events-per-variable ≥ 10).

R Implementation

```r
d <- mtcars
d$am <- factor(d$am)

fit_probit <- glm(am ~ wt + hp, data = d,
                  family = binomial(link = "probit"))
summary(fit_probit)

# Compare with logit
fit_logit <- glm(am ~ wt + hp, data = d, family = binomial)
```

```r
# Predictions should be similar
pred_p <- predict(fit_probit, type = "response")
pred_l <- predict(fit_logit, type = "response")
cor(pred_p, pred_l)
```

Output & Results

`summary(fit_probit)` returns coefficients on the probit scale with Wald z-tests. The correlation between probit and logit predicted probabilities is typically above 0.999, confirming that the two methods are interchangeable for prediction purposes.

Interpretation

A reporting sentence: "Probit regression estimated that each 1000-lb increase in vehicle weight reduced the latent propensity for manual transmission by 1.8 SDs (probit coefficient -1.8, 95 % CI -3.0 to -0.7, $p = 0.002$); the average marginal effect on the probability scale was -0.27 per 1000 lbs (95 % CI -0.42 to -0.13, computed via `margins`)." Always pair probit coefficients with average marginal effects when communicating to non-statistical audiences.

Practical Tips

- Probit and logistic give essentially identical predicted probabilities; choose based on convention or latent-variable interpretation, not on prediction accuracy.
- Probit coefficients lack the clean odds-ratio interpretation of logistic; report average marginal effects (`margins::margins`, `marginaleffects::avg_predictions`) for clinical communication.
- For multilevel or Bayesian binary models, probit links are sometimes computationally simpler than logistic; `brms` and `Stan` handle both.
- The probit-vs-logit choice is almost always a matter of reporting convention rather than statistical superiority.
- For ordinal extensions, probit gives the cumulative-probit (or threshold) model; for survival analysis, the probit link appears in some accelerated-failure-time formulations.
- Predictions from both models should agree to within 1–2 percentage points across the full range of fitted probabilities; large discrepancies signal model misspecification.

R Packages Used

Base R `glm()` with `family = binomial(link = "probit")`; `margins::margins()` and `marginaleffects::avg_predictions()` for average marginal effects; `mfx::probitmfx()` for direct marginal-effect output; `mvProbit` for multivariate probit when multiple binary outcomes are jointly modelled.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring

- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI